

TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB Scrambler

Formant Scramble

Rearranges the formant envelope and fine spectral structure inside a frequency range you choose, turning voices, drums, and effects into metallic, encrypted, radio-like textures.

31 spectral bands

LOW and HIGH range

One AMOUNT control

12 factory presets



Contents

- | | | | |
|-----------|-------------|-----------|--------------------------|
| 01 | Overview | 06 | Recipes |
| 02 | Quick start | 07 | Tips and troubleshooting |
| 03 | Interface | 08 | Parameter reference |
| 04 | Controls | 09 | Version history |
| 05 | Presets | 10 | Support and license |

01 Overview

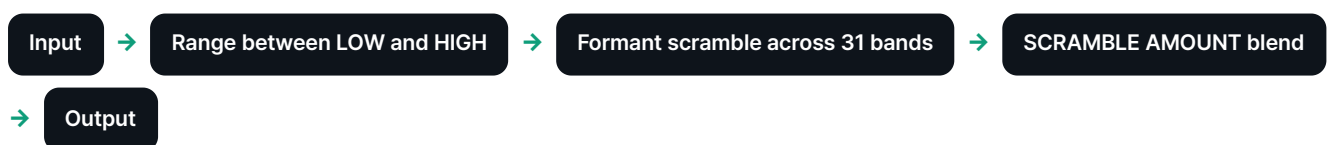
TB Scrambler breaks the recognisable shape of a sound inside a frequency range you choose. Between LOW and HIGH, the formant envelope and fine spectral structure are rearranged; outside that range, the sound passes through unchanged.

Because the low end, rhythm, and overall energy are preserved, vocals, drum loops, and effects can turn metallic, encrypted, or radio-like and still sit in a mix. **SCRAMBLE AMOUNT** moves continuously from the untouched sound to full displacement.

Key features

- **LOW** and **HIGH** set the scrambled range anywhere from 20 Hz to 20 kHz
- Crossed points are reordered, so LOW and HIGH can be set in either order
- Frequencies outside the range pass through unchanged
- **SCRAMBLE AMOUNT** blends from dry (0%) to fully scrambled (100%)
- A repeatable permutation: the same settings always scramble the same way
- 12 factory presets in three groups, user presets, undo, and redo

Signal flow



Frequencies outside the range bypass the scramble.

About this manual

This manual describes TB Scrambler 2.3.1 for Windows (VST3). The current macOS build is version 1.1.4 and may differ from the descriptions here.

02 Quick start

- 1 Insert TB Scrambler on a vocal, drum loop, or effect. The defaults are **LOW** 300 Hz, **HIGH** 4.5 kHz, and **SCRAMBLE AMOUNT** 50%.

- 2 Set **LOW** and **HIGH** around the part of the spectrum you want to change. Everything outside that range stays as it is.
- 3 Raise **SCRAMBLE AMOUNT** until the sound is as unrecognisable as you want.
- 4 Try the factory presets: GENTLE OFFSET for subtle colour, BROKEN SPEECH for garbled voices, and TOTAL DISLOCATION for complete transformation.

03 Interface



TB Scrambler 2.3 interface (portfolio image).

Area	What it does
Preset bar	Previous and next preset, the preset list (factory preset groups and LOAD USER PRESET...; CUSTOM appears after a change), Save, Undo, and Redo.
LOW , HIGH	The two edges of the scrambled range.
SCRAMBLE AMOUNT	How strongly the range is scrambled.

04 Controls

Control	Range · default	What it does
LOW	20 Hz to 20 kHz · 300 Hz	First scramble point. Frequencies below the range pass through unchanged.
HIGH	20 Hz to 20 kHz · 4.5 kHz	Second scramble point. Frequencies above the range pass through unchanged.
SCRAMBLE AMOUNT	0 to 100% · 50%	Dry/wet blend. 0% leaves the sound dry; 100% is fully formant-scrambled.

If LOW is set above HIGH, the two points are reordered, so the range always runs from the lower value to the higher one.

How the scramble works

TB Scrambler divides the spectrum into 31 bands. In the bands between LOW and HIGH, the formant envelope and fine spectral structure are moved around with a fixed permutation, so the same settings always give the same result. The low end, rhythm, and overall energy are kept, which is why the result stays usable in a mix.

Wider ranges destroy recognition faster but also blur the original rhythm and tone. For a subtle effect, keep the range narrow and the amount low.

05 Presets

Step through presets with the arrows, or open the list to choose a factory or user preset. LOAD USER PRESET... opens a .tbscrambler file, and Save stores the current settings as one. The name shows CUSTOM after you change a setting. Undo and Redo step through recent changes.

Group	Preset	LOW	HIGH	AMOUNT
Gentle Offset	Init	300 Hz	4.5 kHz	50%
Gentle Offset	Vocal Haze	600 Hz	3 kHz	22%
Gentle Offset	Drum Grit	400 Hz	6 kHz	18%
Gentle Offset	Bass Wobble	120 Hz	1.8 kHz	26%
Broken Speech	Mid Fracture	420 Hz	4.2 kHz	55%
Broken Speech	Radio Scramble	720 Hz	3.2 kHz	68%
Broken Speech	Yolk Crumble	180 Hz	6.5 kHz	44%
Broken Speech	Formant Flip	500 Hz	3.6 kHz	82%
Total Dislocation	Wide Disorder	90 Hz	11 kHz	78%
Total Dislocation	Alien Broadcast	250 Hz	8 kHz	92%
Total Dislocation	Glass Shatter	1.2 kHz	12 kHz	100%
Total Dislocation	Total Dislocation	20 Hz	20 kHz	100%

06 Recipes

Starting points for common jobs. Adjust them by ear.

Disguised voice

A voice that keeps its rhythm but loses its identity.

Preset	Radio Scramble
Range	About 700 Hz to 3.2 kHz
AMOUNT	60 to 70%

- 1 Keep the range inside the speech band so the timing stays readable.

Drum loop grit

Metallic colour on a loop without losing the groove.

Preset	Drum Grit
Range	400 Hz to 6 kHz
AMOUNT	15 to 30%

- 1 Raise AMOUNT slowly; small values already change the colour.

Alien transmission

A completely transformed, broadcast-like voice.

Preset	Alien Broadcast
Range	250 Hz to 8 kHz
AMOUNT	90 to 100%

- 1 Follow with a band-pass EQ for an even narrower radio sound.

Bass that keeps its weight

Movement in the upper harmonics of a bass.

Preset	Bass Wobble
Range	120 Hz to 1.8 kHz
AMOUNT	20 to 30%

- 1 Raise LOW above the fundamental if the bass loses weight.

07 Tips and troubleshooting

Symptom	What to check
Nothing changes.	SCRAMBLE AMOUNT is at 0%, or the range covers a part of the spectrum with little energy. Move LOW and HIGH.
The low end changed or disappeared.	Raise LOW above the bass so it stays outside the range.
The effect is too strong.	Lower SCRAMBLE AMOUNT or narrow the range.
The words are no longer understandable.	Use a Gentle Offset preset or keep AMOUNT below about 30%.
A saved preset does not load.	The file must be a TB Scrambler .tbscrambler preset.

08 Parameter reference

Panel label	Host parameter	Range	Default
LOW	Low	20 Hz to 20 kHz	300 Hz
HIGH	High	20 Hz to 20 kHz	4.5 kHz
SCRAMBLE AMOUNT	Scramble Amount	0 to 100%	50%

09 Version history

Version	Changes
2.3.1 · 2026-08-21	Brought the current LOW, HIGH, and AMOUNT scrambler design and the latest processing build to Windows VST3, and aligned the catalog, VST3, and installer versions at 2.3.1.
2.3.0 · 2026-07-30	Added spacing between the upper-right controls and the effect logo so header elements no longer overlap, and normalised the logo size for small window sizes.

10 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

TB Audio plug-ins are free for personal and commercial work under the TB Audio Plug-ins Freeware License. If you release or monetize work made with them, add a credit such as **Made with TB Audio Plug-ins by Team Impulse Impact**. Selling or bundling the plug-ins is not allowed.

If the plug-ins help your music, you can support development at ko-fi.com/teamimpulseimpact.